AQUÍ VA EL TÍTULO DEL ARTÍCULO

JUAN CARLOS MONTER CORTÉS AND LUIS ÁNGEL ZALDÍVAR CORICHI

ABSTRACT. Aquí va el resumen

CONTENTS

Todo list 1
1. Introduction 1
2. Preliminaries 2
3. Separation properties induced by compact quotients 4
3.1. Comparison with Classical Separation Axioms 8
3.2. Inheritance and preservation properties 15
4. Interaction among compact-quotient separation properties 21
5. Relationship with the patch assembly 27
References 29

TODO LIST

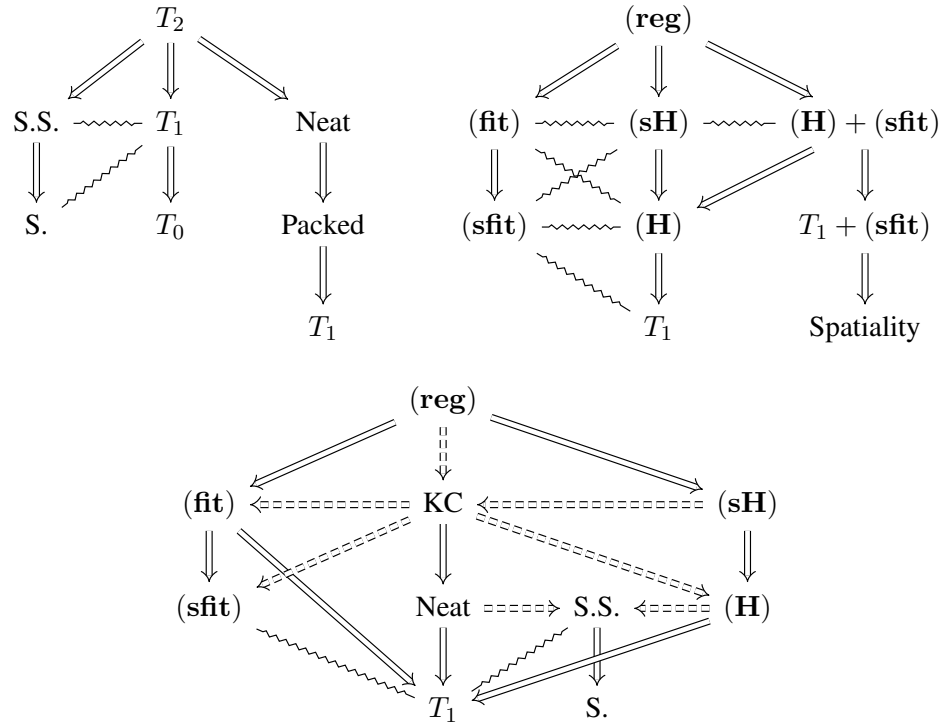
Completar la introducción. 1
DONE: hay que modificar el formato bastante, hay muchas separaciones en las demostraciones eso se ve raro tambien te suegiero que agrres un paper que te guste y veas su estructura de como escrbiben las cosas . . 1
DONE: Combianar secciones 2 y 3 4
Existe alguna propiedad P tal que (H)+P implica staked 23
T1+S.+P implica (H)? 23
DONE: Relacionar S. y S.S. con fit y sfit 23
Relacionar el trabajo con lo presentado por Guram y Sebastian 23
DONE: Ejemplo de neat no KC 23
DONE: Comprobar si neat implica stacked 23
DONE: KC implica S.S.? 23
Completar esta prueba 28
DONE: Considerar incluir información sobre los parches 28

1. INTRODUCTION

[3, 11]

Completar la introducción.

DONE: hay que modificar el formato bastante, hay muchas separaciones en las demostra-



2. PRELIMINARIES

We recall the basic notions and notation concerning frames, nuclei, and open filters; for general background, see the standard references on point-free topology ([5, 8, 14]).

A *frame* is a complete lattice A satisfying

$$a \wedge \bigvee X = \bigvee \{a \wedge x \mid x \in X\}.$$

A frame morphism preserves finite meets and arbitrary joins. We write Frm for the category of frames. The Heyting implication on A is characterized by

$$x \wedge a \leq b \quad \Longleftrightarrow \quad x \leq (a \succ b).$$

Every frame morphism $f\colon A \to B$ has a right adjoint $f_* : B \to A$, characterized by $f(a) \leq b$ if and only if $a \leq f_*(b)$.

A *nucleus* on A is a map $j\colon A \to A$ satisfying

$$x \leq j(x), \qquad j(j(x)) = j(x), \qquad j(x \wedge y) = j(x) \wedge j(y).$$

If j is a nucleus, then the fixed-point set $A_j = \{x \in A \mid j(x) = x\}$ is a frame, with joins given by $\bigvee_{A_j} X = j(\bigvee_A X)$. The map $j^*\colon A \to A_j$, $j^*(x) = j(x)$, is a surjective frame morphism, and every quotient arises in this way. We denote by NA the frame of nuclei on A ([5, Proposition II.2.5]). For $a \in A$, put

$$u_a(x) = a \vee x, \qquad v_a(x) = a \succ x, \qquad w_a(x) = (x \succ a) \succ a.$$

The nuclei u_a and v_a determine the closed and open quotients, respectively.

Every frame morphism $f\colon A \to B$ induces an adjunction in NA. For $k \in NB$, its *lifting* along f is

$$N(f)_*(k) = f_* \circ k \circ f, \tag{1}$$

whereas, for $j \in NA$, its *pushforward* along f is

$$N(f)(j) = \bigvee \{u_{f(j(a))} \wedge v_{f(a)} \mid a \in A\}. \tag{2}$$

Thus, for $j \in NA$ and $k \in NB$,

$$N(f)(j) \leq k \quad \iff \quad j \leq N(f)_*(k).$$

An element $p \in A$ is $\wedge$-*irreducible* if $p \neq 1$ and $x \wedge y \leq p$ implies $x \leq p$ or $y \leq p$. The set $\mathrm{pt}(A)$ of such elements is endowed with the topology generated by

$$\mathcal{U}(a) = \{p \in \mathrm{pt}(A) \mid a \not\leq p\}, \qquad a \in A.$$

The assignment $\mathcal{U}\colon A \to \mathcal{O}(\mathrm{pt}(A))$ is a surjective frame morphism, called the *spatial reflection*. A frame is *spatial* when $\mathcal{U}$ is an isomorphism. Points correspond bijectively to completely prime filters through $F_p = \{a \in A \mid a \not\leq p\}$.

A filter F on A is *open* if, for every directed $D \subseteq A$,

$$\bigvee D \in F \quad \Longrightarrow \quad D \cap F \neq \varnothing.$$

We write $A^\wedge$ for the collection of open filters.

Proposition 2.1. *Let $f = f^*\colon A \to B$ be a frame morphism. If $G \in B^\wedge$, then $f_*G \in A^\wedge$ and*

$$f_*G = \{a \in A \mid f(a) \in G\}. \tag{3}$$

Every nucleus j determines its *admissible filter*

$$\nabla(j) = \{a \in A \mid j(a) = 1\}.$$

Given a filter F, the least nucleus admitting F is

$$v_F = f_F^\infty = \bigvee_{a \in F} v_a, \qquad f_F(x) = \bigvee_{a \in F} (a \succ x), \tag{4}$$

where the join defining v_F is taken in NA. For $F \in A^\wedge$, one has $\nabla(v_F) = F$. Thus, open filters correspond bijectively to fitted nuclei whose admissible filters are open ([2, Lemmas 4.2 and 4.3]).

For a topological space S and $E \subseteq S$,

$$E^\circ, \quad \overline{E}, \quad E',$$

denote the interior, closure, and complement of E, respectively. Each subset E induces a nucleus

$$[E] : \mathcal{O}S \longrightarrow \mathcal{O}S, \qquad [E](U) = (E \cup U)^\circ.$$

A nucleus of this form is called *spatially induced*.

Writing $\mathcal{C}S$ for the lattice of closed subsets of S, E induces an idempotent *derivative*

$$\langle E\rangle : \mathcal{C}S \to \mathcal{C}S, \qquad \langle E\rangle(X) = \overline{E \cap X}.$$

It is dual to $[E']$ under complementation:

$$\langle E\rangle(X) = \big([E'](X')\big)'.$$

For $S = \mathrm{pt}(A)$, the Hofmann–Mislove Theorem ([4, Theorem 2.16]), identifies $A^\wedge$ with the compact saturated subsets of S ($\mathcal{Q}S$). Explicitly, the corresponding sets and filters are

$$Q_F = \{p \in \mathrm{pt}(A) \mid (\forall a \in F)\ a \not\leq p\}, \qquad F_Q = \{a \in A \mid Q \subseteq \mathcal{U}(a)\}.$$

When the corresponding open filter or compact saturated subset is clear from the context, we write simply (F, Q) for a pair associated by the Hofmann–Mislove theorem and call it an *HM pair*. If $\mathrm{pt}(A) = \varnothing$, the only open filter is A, corresponding to $Q = \varnothing$ and to the trivial fitted quotient $v_A = \top_{NA}$.

If $Q \in \mathcal{Q}S$, we denote by $\partial_F = \langle Q\rangle$ the associated idempotent derivative; thus

$$\partial_F(X) = \overline{Q \cap X}, \qquad X \in \mathcal{C}S.$$

Recall that the quotient A_j determined by a nucleus j is compact if and only if $\nabla(j)$ is an open filter ([2, Lemma 4.4]). In particular, for each $F \in A^\wedge$, the fitted nucleus v_F determines a compact quotient. However, a compact quotient need not be uniquely determined by its admissible filter.

For $F \in A^\wedge$, let

$$M = \{m \in A \setminus F \mid m \text{ is maximal in } A \setminus F\}. \tag{5}$$

Define

$$w_F(x) = \bigwedge\{m \in M \mid x \leq m\}$$

and the nucleus w_F is the greatest nucleus with admissible filter F, whereas v_F is the least one. Consequently,

$$\{j \in NA \mid \nabla(j) = F\} = [v_F, w_F].$$

We call this the *admissibility interval* associated with F. Evaluating at 0 yields

$$[v_F(0), w_F(0)] \subseteq A.$$

Let $Q \in \mathcal{Q}S$, put $G = \{U \in \mathcal{O}S \mid Q \subseteq U\}$. Then

$$\nabla([Q']) = G, \qquad v_G \leq [Q'] \leq w_G.$$

More generally, let $F \in A^\wedge$, $Q \in \mathcal{Q}S$, and put $G \in \mathcal{O}S^\wedge$, we call (F, Q, G) an *HM triple*.

DONE: Combianar secciones 2 y 3

3. Separation properties induced by compact quotients

The notions studied in this section are motivated by the work of Sexton and Simmons on point-sensitive and point-free patch constructions [10, 11, 12, 13]. Our aim is to identify nuclei associated with open filters that determine compact closed quotients, and to investigate the separation properties arising from this requirement. We begin by recalling Sexton's notion of neatness.

Definition 3.1. Let A be a frame, let $F \in A^\wedge$ and $\alpha \in \mathrm{Ord}$. We say that:

- F is α-neat, if
$$x \vee d = 1, \quad \text{for every } x \in F,$$
where $d = f_F^\alpha(0)$.
- A is α-neat if every $F \in A^\wedge$ is α-neat.
- A is neat if it is α-neat for some α.

Let us record a useful consequence of neatness. Suppose that A is α-neat and let $F \in A^\wedge$. Put $d = f_F^\alpha(0)$, then $x \vee d = 1$ for every $x \in F$, we have $F \subseteq \nabla(u_d)$. Thus, $f_F \leq u_d$, and consequently
$$v_F = f_F^\infty \leq u_d.$$
On the other hand,
$$d = f_F^\alpha(0) \leq v_F(0),$$
so that
$$u_d \leq u_{v_F(0)} \leq v_F.$$
Therefore,
$$v_F = f_F^\infty = u_d.$$
In particular, every fitted nucleus associated with an open filter is closed, and hence every such compact quotient is simultaneously closed.

Sexton showed that a topological space is neat if and only if it is both packed and stacked [10, Theorem 6.2]. Motivated by this spatial characterization, we now introduce point-free counterparts of stackedness and strong stackedness.

Let S be a topological space and let ($G \in \mathcal{O}S^\wedge$, $Q \in \mathcal{Q}S$) be an HM pair. Recall that S is stacked if
$$v_G(\varnothing) = (Q')^\circ$$
for every such pair (G, Q), and strongly stacked if
$$v_G = [Q']$$
for every such pair.

Denote by
$$\mathcal{U}_* : \mathcal{O}(\mathrm{pt}(A)) \to A$$
the right adjoint of the spatial reflection.

Proposition 3.2. *For every $k \in [v_G, w_G]$, the lifiting*
$$\mathcal{U}_* \circ k \circ \mathcal{U}$$
is a nucleus on A with admissible filter F.

Proof. The composite $\mathcal{U}_* \circ k \circ \mathcal{U}$ is a nucleus on A. Moreover, since $\nabla(k) = G$ and $\mathcal{U}_*(V) = 1$ if and only if $V = S$, we obtain
$$\begin{aligned}\nabla(\mathcal{U}_* \circ k \circ \mathcal{U}) &= \mathcal{U}^{-1}[\nabla(k)] \\ &= \mathcal{U}^{-1}[G] = F.\end{aligned}$$
Hence $\mathcal{U}_* \circ k \circ \mathcal{U}$ belongs to the admissibility interval $[v_F, w_F]$. □

Hence Proposition 3.2 gives the map

$$\Phi\colon [v_G, w_G] \longrightarrow [v_F, w_F], \qquad \Phi(k) = \mathcal{U}_* \circ k \circ \mathcal{U},$$

and, in particular,

$$v_F \leq \Phi(k) \leq w_F.$$

Finally, when $A = \mathcal{O}S$, identifying S with $\mathrm{pt}(A)$, the spatial reflection is an isomorphism.

Corollary 3.3. *If $A = \mathcal{O}S$, then $\mathcal{U}_* \circ [E] \circ \mathcal{U} = [E]$ for every $E \subseteq S$.*

Proof. Since $\mathcal{U}$ is an isomorphism, $\mathcal{U}_* = \mathcal{U}^{-1}$. Thus the lifting Φ reduces to the identity under the canonical identification, and hence $\Phi([E]) = [E]$. □

We now return to the general setting. Let A be a frame, $S = \mathrm{pt}(A)$, and let (F, Q, G) be an H–M triple.

Proposition 3.4. *Let (F, Q, G) be an H–M triple. Then $\Phi(w_G) = w_F$, that is,*

$$w_F = \mathcal{U}_* \circ w_G \circ \mathcal{U}.$$

Proof. By Proposition 3.2, we have

$$\Phi(w_G) = \mathcal{U}_* \circ w_G \circ \mathcal{U} \leq w_F.$$

For the reverse inequality, by adjunction it suffices to show that

$$\mathcal{U} \circ w_F \leq w_G \circ \mathcal{U}.$$

Since $w_G = [M']$, for every $x \in A$ we have

$$w_G(\mathcal{U}(x)) = [M'](\mathcal{U}(x)) = (M' \cup \mathcal{U}(x))^\circ.$$

Thus, it is enough to prove that

$$\mathcal{U}(w_F(x)) \subseteq M' \cup \mathcal{U}(x).$$

Suppose that $y \in \mathcal{U}(w_F(x))$ and $y \notin M'$. Since $y \notin M'$, we have $y \in M$. Moreover, $y \in \mathcal{U}(w_F(x))$ implies $w_F(x) \not\leq y$. But

$$w_F(x) = \bigwedge \{m \in M \mid x \leq m\}.$$

Hence $x \not\leq y$, since otherwise $w_F(x) \leq y$, a contradiction. Therefore $y \in \mathcal{U}(x)$, proving the desired inclusion. Consequently, $\Phi(w_G) = w_F$. □

Proposition 3.4 shows that Φ preserves the upper elements of the corresponding admissibility intervals. Since $[Q']$ is the spatially induced nucleus appearing in Sexton's definitions, the nucleus

$$\Phi([Q']) = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}$$

provides a natural point-free counterpart.

Definition 3.5. Let A be a frame and (F, Q) an H–M pair. We say that:

- A is *FQ-stacked* if

$$(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) = v_F(0).$$

- A is *FQ-strongly stacked* if

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = v_F.$$

The frame A is *stacked* (respectively, *strongly stacked*) if it is FQ-stacked (respectively, FQ-strongly stacked) for every H–M pair (F, Q).

Clearly, every strongly stacked frame is stacked. Let us look that these properties are conservative. We first establish the relation between the transfinite iterations defining the fitted nuclei on a space and on its frame of open sets.

Lemma 3.6. *Let $\alpha \in \mathrm{Ord}$ and $X \in CS$. Then*

$$\partial_F^\alpha(X) = \big(f_F^\alpha(X')\big)'.$$

In particular, for every $U \in \mathcal{O}S$,

$$\big(\partial_F^\infty(U')\big)' = v_F(U).$$

Proof. We proceed by transfinite induction on α. For $\alpha = 0$, the identity follows from $\partial_F^0 = \mathrm{id} = f_F^0$.

Suppose that the identity holds for α. Then

$$\begin{aligned}\partial_F^{\alpha+1}(X) &= \partial_F(\partial_F^\alpha(X)) \\ &= \big(f_F((\partial_F^\alpha(X))')\big)' \\ &= \big(f_F(f_F^\alpha(X'))\big)' \\ &= \big(f_F^{\alpha+1}(X')\big)'.\end{aligned}$$

If λ is a limit ordinal, then

$$\begin{aligned}\partial_F^\lambda(X) &= \bigcap_{\alpha<\lambda} \partial_F^\alpha(X) \\ &= \bigcap_{\alpha<\lambda} (f_F^\alpha(X'))' \\ &= \left(\bigcup_{\alpha<\lambda} f_F^\alpha(X')\right)' \\ &= \big(f_F^\lambda(X')\big)'.\end{aligned}$$

Thus, the identity holds for every ordinal α. Taking the stable value of the iterations yields the last assertion. □

Proposition 3.7. *The notions of stackedness and strong stackedness are conservative.*

Proof. Let S be a topological space and put $A = \mathcal{O}S$. For every H–M pair (F, Q), we have $v_F = [Q']$ whenever S is strongly stacked. By Lemma 3.3,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = [Q'] = v_F.$$

Thus, A is strongly stacked.

Conversely, if A is strongly stacked, then

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U} = [Q']$$

for every H–M pair (F, Q), and hence S is strongly stacked.

Now suppose that S is stacked. By definition,

$$Q = \partial_F^\infty(S).$$

Consequently,

$$Q' = (\partial_F^\infty(S))' = v_F(\varnothing),$$

where the last equality follows from Lemma 3.6. By Lemma 3.3,

$$\begin{aligned}(U_* \circ [Q'] \circ U)(\varnothing) &= [Q'](\varnothing) \\ &= (Q')^\circ \\ &= Q' \\ &= v_F(\varnothing).\end{aligned}$$

Therefore, A is stacked.

Conversely, if A is stacked, then

$$v_F(\varnothing) = (U_* \circ [Q'] \circ U)(\varnothing) = Q'.$$

By Lemma 3.6,

$$(\partial_F^\infty(S))' = Q',$$

and hence $\partial_F^\infty(S) = Q$. Thus, S is stacked. □

Neatness requires every fitted nucleus associated with an open filter to determine a closed compact quotient. A natural extension is to impose this condition on every nucleus admitting an open filter. This leads to a point-free counterpart of Wilansky's classical KC property [17].

Definition 3.8. Let A be a frame. We say that A is *KC* if every compact quotient of A is closed.

In the spatial setting, Wilansky's KC property coincides with Sexton's notion of packedness.

Whether analogous relationships hold in the point-free setting will be addressed in Section 4. We first examine how these properties interact with classical separation axioms.

3.1. **Comparison with Classical Separation Axioms.** Although stackedness and strong stackedness are conservative, neither property is comparable with the T_1 separation axiom, as the following examples show.

Example 3.9.

(1) The frame $A = [0, 1]$, studied in Example 5.1, is strongly stacked but not T_1. Indeed, for every open filter $F_r = (r, 1]$, $0 \leq r < 1$, the corresponding compact saturated set is $Q_r = [0, r]$, and

$$v_{F_r}(x) = \begin{cases} x, & x \leq r, \\ 1, & x > r, \end{cases} = (\mathcal{U}_* \circ [Q_r'] \circ \mathcal{U})(x).$$

(2) The Hausdorff frame of Paseka and Šmarda [7], considered in Example 3.18, is T_1 but not stacked, and hence not strongly stacked.

Recall that every T_1 stacked space is sober and that, in the strongly stacked case, $w_G = v_G$. We now investigate analogous phenomena for frames.

Proposition 3.10. *Let A be a frame. If*

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$, then every point of A is maximal in $\operatorname{pt} A$.

Proof. Let $p, q \in \operatorname{pt} A$ with $p \leq q$, and consider the open filter

$$F_q = \{x \in A \mid x \not\leq q\}.$$

By the definition of w_{F_q},

$$w_{F_q} = v_q, \qquad \text{so} \qquad w_{F_q}(0) = q.$$

By hypothesis,

$$q = v_{F_q}(0) = \bigvee_{x \in F_q} \neg x.$$

For every $x \in F_q$, we have $x \not\leq q$, and hence $x \not\leq p$. Since p is a point,

$$x \wedge \neg x = 0 \leq p \quad \Longrightarrow \quad \neg x \leq p.$$

Consequently,

$$q = \bigvee_{x \in F_q} \neg x \leq p.$$

Thus $p = q$, proving that every point of A is maximal. □

If A is a frame T_1, then $\operatorname{pt}(A)$ is T_1 and, the set M in (5), satisfies $M = Q$.

Proposition 3.11. *Let A be a T_1 frame. Then:*

(1) *A is stacked if and only if*

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$.

(2) *A is strongly stacked if and only if*

$$v_F = w_F$$

for every $F \in A^\wedge$.

Proof. Let (F, Q) be an H–M pair. Since A is T_1, we have $M = Q$, and hence $w_G = [Q']$. By Proposition 3.4,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = w_F.$$

Both assertions now follow directly from the definitions of stackedness and strong stackedness. □

Remark 3.12. In Proposition 3.11, it suffices to assume that $v_F(0) = w_F(0)$ (respectively, $v_F = w_F$) for every $F \in A^\wedge$ to conclude that A is stacked (respectively, strongly stacked), without requiring A to be T_1.

We now consider conditions under which the corresponding identities imply the T_1 separation property.

Corollary 3.13. *Let A be a frame such that*

$$\mathcal{V}(a) = \{p \in \operatorname{pt} A \mid a \leq p\} \neq \varnothing$$

for every $a \neq 1$. If

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$, then A is T_1. In particular, the same conclusion holds if $v_F = w_F$ for every $F \in A^\wedge$.

Proof. By Proposition 3.10, every point of A is maximal in $\operatorname{pt} A$. Suppose that there exist $p \in \operatorname{pt} A$ and $a \in A$ such that $p < a < 1$. By hypothesis, there is $q \in \operatorname{pt} A$ with $a \leq q$. Hence

$$p < a \leq q,$$

contradicting the maximality of p in $\operatorname{pt} A$. Therefore, every point of A is maximal in A, and hence A is T_1. The last assertion follows immediately from $v_F = w_F$. □

While the strongly stacked condition yields the collapse of every admissibility interval, the stacked condition already guarantees this phenomenon for those associated with completely prime filters.

Corollary 3.14. *Let A be a T_1 stacked frame. Then $[v_{F_p}, w_{F_p}] = \{u_p\}$, for every $p \in \operatorname{pt} A$.*

Proof. Let $p \in \operatorname{pt} A$. Since A is T_1 and stacked, by Proposition 3.11 we have

$$v_{F_p}(0) = w_{F_p}(0) = p.$$

Moreover, since p is maximal in A, we have

$$w_{F_p} = u_p.$$

For every nucleus j on A, one has $u_{j(0)} \leq j$. Applying this to v_{F_p}, we obtain

$$u_p = u_{v_{F_p}(0)} \leq v_{F_p} \leq w_{F_p} = u_p.$$

Consequently,

$$[v_{F_p}, w_{F_p}] = \{u_p\}.$$

□

Notice that the additional hypothesis in Corollary 3.13 is automatically satisfied whenever A is spatial. Indeed, if $a < 1$, spatiality ensures the existence of $p \in \operatorname{pt} A$ such that $a \leq p$. Thus, in the spatial setting, our results recover the classical fact that every T_1 stacked space is sober.

The point-free situation is substantially different: even the stronger combination of T_1 and strong stackedness does not imply spatiality, as the following example shows.

Example 3.15. Let B be an atomless complete Boolean algebra and consider the frame

$$A = B \times 2,$$

where $2 = \{0 < 1\}$. Since B has no points and points of a finite product of frames correspond to points of its factors, we have

$$\operatorname{pt} A = \{p\}, \qquad p = (1, 0).$$

Clearly, p is maximal in A, so A is T_1. However, A is not spatial, since for distinct $a, b \in B$,

$$(a, 0) \neq (b, 0), \qquad \mathcal{U}(a, 0) = \mathcal{U}(b, 0) = \varnothing.$$

We now show that A is strongly stacked. Since B is an atomless complete Boolean algebra, its only compact element is 0. Thus,

$$K(A) = K(B) \times K(2) = \{(0, 0), (0, 1)\}.$$

As every open filter of a complete Boolean algebra is principal and generated by a compact element, it follows that

$$A^\wedge = \{A, F_p\}, \qquad F_p = B \times \{1\}.$$

Since $F_p = {\uparrow}(0, 1)$ and A is a complete Boolean algebra, we have

$$v_{F_p} = v_{(0,1)} = u_p.$$

Moreover, p is the unique maximal element of $A \setminus F_p$, so

$$w_{F_p} = u_p.$$

Consequently,

$$v_{F_p} = w_{F_p} = u_p.$$

For $F = A$,

$$v_A = \operatorname{tp} = w_A.$$

Then,

$$v_F = w_F \qquad \text{for every } F \in A^\wedge.$$

By Proposition 3.11, A is strongly stacked.

The preceding example yields the following result,

Theorem 3.16. *There exist non-spatial T_1 strongly stacked frames. In particular, there exist non-spatial T_1 stacked frames.*

Recall that $A^\wedge$ carries the topology generated by the basic open sets

$$D(x) = \{F \in A^\wedge \mid x \in F\}, \qquad x \in A.$$

Moreover, the points of the Vietoris modification VA (view [6]), are represented by pairs

$$(F, a), \qquad F \in A^\wedge, \quad a \in [v_F(0), w_F(0)].$$

Consider the continuous surjection

$$\pi : \operatorname{pt}(VA) \longrightarrow A^\wedge, \qquad (F, a) \longmapsto F.$$

It induces the frame morphism

$$\mathcal{O}(\pi) = \pi^{-1} \colon \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\mathrm{pt}(VA)).$$

Recall also that the subbasic open sets of $\mathrm{pt}(VA)$ are determined by

$$(F, a) \in \Box x \iff x \in F, \qquad (F, a) \in \Diamond x \iff x \not\leq a. \tag{6}$$

Particularly,

$$a \in [F](x) \iff x \in F \quad \text{and} \quad a \in \langle F\rangle(x) \iff x \not\leq a. \tag{7}$$

Theorem 3.17. *Let A be a frame, and consider the continuous surjection*

$$\pi \colon \mathrm{pt}(VA) \longrightarrow A^\wedge, \qquad (F, a) \longmapsto F.$$

Then the following statements hold:

(1) *If A is T_1 and stacked, then*

$$\mathcal{O}(\pi) = \pi^{-1} \colon \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\mathrm{pt}(VA))$$

is an isomorphism.

(2) *If $\mathcal{O}(\pi)$ is an isomorphism, then A is stacked.*

(3) *If $\mathcal{O}(\pi)$ is an isomorphism and*

$$\mathcal{V}(a) = \{p \in \mathrm{pt}\, A \mid a \leq p\} \neq \varnothing$$

for every $a < 1$, then A is T_1.

Proof.

(1) Suppose that A is T_1 and stacked. By Proposition 3.11,

$$v_F(0) = w_F(0) \qquad (F \in A^\wedge).$$

Thus, every fiber of π is a singleton, so π is bijective.

It remains to show that π^{-1} is continuous. Recall that the subbasic open sets of $\mathrm{pt}(VA)$ are

$$[-](x) = \{(F, a) \mid x \in F\}$$

and

$$\langle - \rangle(x) = \{(F, a) \mid x \not\leq a\}.$$

For the first family,

$$\pi([-](x)) = D(x).$$

For the second, since $a = v_F(0)$ and $v_F(0) = w_F(0)$, we have

$$\begin{aligned} (F, a) \in \langle - \rangle(x) &\iff x \not\leq v_F(0) \\ &\iff x \not\leq w_F(0) \\ &\iff x \in F. \end{aligned}$$

Hence,

$$\pi(\langle - \rangle(x)) = D(x).$$

Therefore, π^{-1} is continuous, and π is a homeomorphism. Consequently, $\mathcal{O}(\pi)$ is an isomorphism.

(2) Suppose that $\mathcal{O}(\pi)$ is an isomorphism. Then π is injective. Since both
$$(F, v_F(0)), \qquad (F, w_F(0))$$
belong to $\mathrm{pt}(VA)$ and have the same image under π, we obtain
$$v_F(0) = w_F(0)$$
for every $F \in A^\wedge$. By Remark 3.12, A is stacked.

(3) Suppose additionally that
$$\mathcal{V}(a) \neq \varnothing \qquad (a < 1).$$
By part (2),
$$v_F(0) = w_F(0) \qquad (F \in A^\wedge).$$
By Corollary 3.13, it follows that A is T_1.

□

We now compare stackedness with the stronger separation property (H). As shown in [7], (H) implies T_1. However, (H) does not imply stackedness.

Example 3.18. Let $I = [0, 1]$ with its usual topology and put $A = \mathcal{O}(I)$. Following the construction of Paseka and Šmarda [7, Proposition 2.3], consider
$$\widetilde{A} = \{(u, v) \in A \times A_{\neg\neg} \mid u \leq v\},$$
where $A_{\neg\neg}$ is the Boolean algebra of regular elements of A.

By [7, Propositions 2.3 and 2.4], $\widetilde{A}$ is a compact frame satisfying (H), with
$$\mathrm{pt}\,\widetilde{A} = \{(I \setminus \{x\}, I) \mid x \in I\}.$$

Since $\widetilde{A}$ is compact, $F = {\uparrow}\, 1 = \{1\}$ is an open filter. Let (F, Q) be its corresponding H–M pair. As $1 \notin \mathrm{pt}\,\widetilde{A}$, we have $Q = \mathrm{pt}\,\widetilde{A}$.

Now $f_F = v_1 = \mathrm{id}_{\widetilde{A}}$, and hence
$$v_F(0) = 0_{\widetilde{A}} = (\varnothing, \varnothing).$$
On the other hand,
$$\begin{aligned}(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) &= \bigwedge Q \\ &= \bigwedge \mathrm{pt}\,\widetilde{A} \\ &= (\varnothing, I).\end{aligned}$$
Thus,
$$v_F(0) \neq (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0),$$
so $\widetilde{A}$ is not stacked.

We next consider a stronger point-free version of the Hausdorff separation axiom. A frame A is said to be strongly Hausdorff, or (sH), if its codiagonal
$$\nabla_A : A \oplus A \longrightarrow A$$
is a closed quotient.

Unlike (H), this condition imposes a strong restriction on compact quotients, as the following result shows.

Proposition 3.19. *Every strongly Hausdorff frame is KC.*

Proof. This follows directly from [9, Theorem 9.2.1], which states that every compact quotient of a strongly Hausdorff frame is closed. □

Proposition 3.20. *Every KC frame is T_1.*

Proof. Let $p \in \operatorname{pt} A$. Since the completely prime filter F_p is open, A_{w_p} is a compact quotient of A. By the KC property, there exists $a \in A$ such that $w_p = u_a$.

Evaluating at 0, we obtain

$$a = u_a(0) = w_p(0) = p,$$

and hence $w_p = u_p$.

If $p < b \leq 1$, then $b \not\leq p$, so

$$1 = w_p(b) = u_p(b) = p \vee b = b.$$

Thus p is maximal in A. Since p was arbitrary, A is T_1. □

We briefly recall two separation properties that will be useful in what follows. A frame A is said to be *fit* if, whenever $a \not\leq b$, there exist $x, y \in A$ such that

$$a \vee x = 1, \qquad y \not\leq b, \qquad x \wedge y \leq b.$$

On the other hand, A is *subfit* if, whenever $a \not\leq b$, there exists $c \in A$ such that

$$a \vee c = 1, \qquad b \vee c \neq 1.$$

Every fit frame is subfit. We denote these properties by $(\mathbf{fit})$ and $(\mathbf{sfit})$, respectively.

Corollary 3.21. *Every subfit KC frame is spatial.*

Proof. By Proposition 3.20, every KC frame is T_1. The conclusion follows from [9, Proposition 2.5.3], since every subfit T_1 frame is spatial. □

Theorem 3.22. *There exists a KC frame which is strongly stacked and not subfit. Consequently,*

$$\mathrm{KC} \not\subseteq (\mathrm{sfit}).$$

Proof. Consider the frame $A = B \times 2$ introduced in Example 3.15. As shown there, A is T_1, strongly stacked, and non-spatial. Moreover,

$$A^\wedge = \{A, F_p\},$$

and

$$v_F = w_F \qquad \text{for every } F \in A^\wedge.$$

Thus, every nucleus in an admissibility interval $[v_F, w_F]$ is closed, so A is KC.

Finally, A cannot be subfit, since every subfit KC frame is spatial by Corollary 3.21. Therefore, A is KC, strongly stacked, and not subfit. □

3.2. **Inheritance and preservation properties.** The notions of stacked and strongly stacked frames are formulated in terms of the admissibility intervals associated with Hofmann–Mislove pairs. A related comparison was established by Simmons [15, Lemma 8.9 and Corollary 8.10]: for every H–M pair (F, Q),

$$\mathcal{U} \circ v_F \leq v_G \circ \mathcal{U}$$

where $G = \nabla(Q)$ and $\mathcal{U}$ is the spatial reflection.

We extend this inequality to arbitrary frame morphisms. Let $h : A \to B$ be a frame morphism, $G \in B^\wedge$, and put

$$F = h_* G \in A^\wedge.$$

The following lemmas establish the comparison between the corresponding prenuclei and their fitted nuclei.

Lemma 3.23. *Let $h : A \to B$ be a frame morphism, $G \in B^\wedge$, and $F = h_* G$. Then*

$$h \circ f_F \leq f_G \circ h.$$

Proof. Since

$$F = \{a \in A \mid h(a) \in G\},$$

for every $a \in F$ and $x \in A$ we have

$$h(v_a(x)) = h(a \succ x) \leq h(a) \succ h(x) = v_{h(a)}(h(x)) \leq f_G(h(x)).$$

Taking joins over $a \in F$ and using join preservation yields

$$h(f_F(x)) \leq f_G(h(x)).$$

□

Lemma 3.24. *Under the hypotheses of the preceding lemma, for every ordinal α,*

$$h \circ f_F^\alpha \leq f_G^\alpha \circ h.$$

In particular,

$$h \circ v_F \leq v_G \circ h.$$

Proof. We argue by transfinite induction on α. The case $\alpha = 0$ is immediate. If the inequality holds for α, then

$$\begin{aligned} h \circ f_F^{\alpha+1} &= h \circ f_F \circ f_F^\alpha \\ &\leq f_G \circ h \circ f_F^\alpha \\ &\leq f_G \circ f_G^\alpha \circ h \\ &= f_G^{\alpha+1} \circ h. \end{aligned}$$

For a limit ordinal λ, preservation of arbitrary joins gives

$$\begin{aligned} h(f_F^\lambda(x)) &= \bigvee_{\alpha<\lambda} h(f_F^\alpha(x)) \\ &\leq \bigvee_{\alpha<\lambda} f_G^\alpha(h(x)) \\ &= f_G^\lambda(h(x)). \end{aligned}$$

Finally, taking an ordinal at which both iterations have stabilized yields

$$h \circ v_F \leq v_G \circ h.$$

□

Proposition 3.25. *If A is a neat frame, then A_j is neat for every $j \in NA$.*

Proof. Suppose that A is α-neat and let $j \in NA$. Let $G \in A_j^\wedge$ and put $F = j_* G \in A^\wedge$. Set

$$\widehat{d} = f_F^\alpha(0), \qquad d = f_G^\alpha(0).$$

By Lemma 3.24, applied to $j^* : A \to A_j$, we have

$$j(\widehat{d}) = j(f_F^\alpha(0)) \leq f_G^\alpha(j(0)) = d.$$

Since j is inflationary,

$$\widehat{d} \leq j(\widehat{d}) \leq d.$$

Now let $y \in G$. Since $y \in A_j$, we have $j(y) = y$, and therefore $y \in F$. As A is α-neat,

$$y \vee \widehat{d} = 1.$$

Consequently,

$$1 = y \vee \widehat{d} \leq y \vee d \leq 1.$$

Thus A_j is α-neat, and hence neat. □

We now turn to coproducts. The comparison established in Lemma 3.24 allows us to prove that neatness is preserved under arbitrary coproducts, without first treating the binary case.

Theorem 3.26. *Neatness is preserved under arbitrary coproducts.*

Proof. Let $\{A_i\}_{i \in I}$ be a family of neat frames, and let α_i be a neatness degree of A_i. Put

$$A = \bigoplus_{i \in I} A_i, \qquad \alpha = \sup_{i \in I} \alpha_i,$$

and denote the canonical frame morphisms by $\iota_i : A_i \to A$.

Let $F \in A^\wedge$ and set

$$F_i = (\iota_i)_*[F] \in A_i^\wedge, \qquad d_i = f_{F_i}^\alpha(0), \qquad d = f_F^\alpha(0).$$

Since every A_i is α-neat,

$$x_i \vee d_i = 1 \qquad (x_i \in F_i).$$

By Lemma 3.24,

$$\iota_i(d_i) \leq f_F^\alpha(\iota_i(0)) \leq d.$$

Hence

$$\bigvee_{i \in I} \iota_i(d_i) \leq d.$$

Now let $x \in F$. Using the standard representation of the coproduct, write

$$x = \bigvee_{k \in K} \bigwedge_{i \in J_k} \iota_i(x_{k,i}),$$

where each $J_k \subseteq I$ is finite. Since F is completely prime, there is some $k \in K$ such that

$$\bigwedge_{i \in J_k} \iota_i(x_{k,i}) \in F.$$

Thus $x_{k,i} \in F_i$ for every $i \in J_k$. Consequently,

$$\begin{aligned} 1 &= \bigwedge_{i \in J_k} \big(\iota_i(x_{k,i}) \vee \iota_i(d_i)\big) \\ &\leq \bigwedge_{i \in J_k} \big(\iota_i(x_{k,i}) \vee d\big) \\ &= \left(\bigwedge_{i \in J_k} \iota_i(x_{k,i})\right) \vee d \\ &\leq x \vee d. \end{aligned}$$

Therefore $x \vee d = 1$ for every $x \in F$, so A is α-neat, and hence neat. □

The preceding results have a categorical consequence. Indeed, the subcategory of neat frames is closed under quotients, by Proposition 3.25, and under arbitrary coproducts, by Proposition 3.26. Hence, the standard criterion for coreflectivity of full subcategories of Frm applies.

Corollary 3.27. *The subcategory of neat frames is a coreflective subcategory of* Frm.

We now examine the inheritance of the stronger properties introduced above. We first show that strong stackedness is inherited by open quotients, and then establish the corresponding result for KC.

Recall that, for a frame morphism $f : A \to B$ and $j \in NA$, the induced nuclei satisfy

$$N(f)(j) = \bigvee_{a \in A} \big(u_{f(j(a))} \wedge v_{f(a)}\big), \qquad N(f_*)(k) = f_* \circ k \circ f.$$

Proposition 3.28. *Let A be a strongly stacked frame and $j \in NA$. Then A_j is strongly stacked.*

Proof. Let $F \in A_j^{\wedge}$, and put

$$G = j_* F, \qquad Q = \mathrm{pt}(A_j) \setminus F, \qquad R = \mathrm{pt}\, A \setminus G.$$

Let $\mathcal{U} \colon A \to \mathcal{O}(\mathrm{pt}\, A)$ and $\mathcal{U}_j \colon A_j \to \mathcal{O}(\mathrm{pt}\, A_j)$ be the corresponding spatial reflections. Set

$$l = N(\mathcal{U}_j)([Q']), \qquad m = N(j^*)(v_G).$$

By the general comparison between the admissibility bounds,

$$v_F \leq l.$$

We prove that

$$l \leq m \leq v_F.$$

Since $G = j_*F$, by Lemma 3.24,

$$j^* \circ v_G \leq v_F \circ j^*.$$

Consequently,

$$\begin{aligned} m &= \bigvee_{x \in A} \big(u_{j^*(v_G(x))} \wedge v_{j^*(x)}\big) \\ &\leq \bigvee_{x \in A} \big(u_{v_F(j^*(x))} \wedge v_{j^*(x)}\big) \leq v_F. \end{aligned}$$

Since A is strongly stacked,

$$v_G = N(\mathcal{U})([R']).$$

Hence

$$m = N(j^*)\big(N(\mathcal{U})([R'])\big).$$

Consider the auxiliary nucleus

$$n = N(\mathcal{U}_j \circ j^*)([Q']).$$

Using the explicit formula for the induced nucleus and the identification $Q = R \cap \mathrm{pt}(A_j)$, one obtains

$$j^*(n(x)) = l(x) \qquad (x \in A_j).$$

Moreover, the description of the right adjoint j_* gives

$$j^*(n(x)) \leq j^*\big(N(\mathcal{U})([R'])(x)\big).$$

Therefore,

$$u_{j^*(n(x))} \wedge v_x \leq u_{j^*(N(\mathcal{U})([R'])(x))} \wedge v_x \leq m.$$

Taking joins over $x \in A_j$ yields $l \leq m$. Thus,

$$v_F \leq l \leq m \leq v_F,$$

and consequently

$$v_F = N(\mathcal{U}_j)([Q']).$$

Therefore A_j is strongly stacked. □

In particular, since every strongly stacked frame is stacked, we also obtain that A_j is stacked.

Proposition 3.29. *Let A be a KC frame and $j \in NA$. Then A_j is KC.*

Proof. Let $k \in N(A_j)$ be such that $(A_j)_k$ is compact. By the correspondence of open filters, the nucleus

$$l = j_* \circ k \circ j^*$$

defines a compact quotient A_l. Since A is KC, $l = u_a$ for some $a \in A$.

For every $x \in A$,

$$a \vee x = l(x) = j_*(k(j^*(x))).$$

Applying j^* and using $j^* \circ j_* = \mathrm{id}_{A_j}$, we obtain

$$j^*(a) \vee j^*(x) = k(j^*(x)).$$

Since j^* is surjective, it follows that

$$k = u_{j^*(a)}.$$

Thus $(A_j)_k$ is closed, and hence A_j is KC. □

Now, we work with coproducts of KC frames.

Proposition 3.30. *Let B be a complete atomless Boolean algebra, and let*

$$D = B \cup \{\omega\},$$

where ω is a new top element. Then D is a compact KC frame which is not regular.

Proof. Write $e = 1_B$. Since ω is a new top element, every cover of ω contains ω. Hence D is compact.

Moreover, since B is a complete atomless Boolean algebra, its only Scott-open filter is B. Consequently,

$$D^\wedge = \{D, \{\omega\}\}.$$

We claim that D is KC. Let $j \in ND$ be a nucleus such that D_j is compact. Then $\nabla(j) \in D^\wedge$.

If $\nabla(j) = D$, then $j(0) = \omega$, and hence $j = u_\omega$.

Otherwise, $\nabla(j) = \{\omega\}$, so $j[B] \subseteq B$. Thus, the restriction $j|_B$ is a nucleus on B. Since every nucleus on a Boolean algebra is closed, there exists $a \in B$ such that

$$j(b) = a \vee b \qquad (b \in B).$$

As $j(\omega) = \omega$, we conclude that $j = u_a$. Therefore, D is KC.

Finally, D is not regular. Indeed, if $a \prec e$, there exists $c \in D$ such that

$$a \wedge c = 0, \qquad e \vee c = \omega.$$

The latter equality forces $c = \omega$, and consequently $a = 0$. Thus,

$$\bigvee \{a \in D : a \prec e\} = 0 \neq e.$$

Hence D is not regular. □

Theorem 3.31. *The class of KC frames is not closed under finite coproducts.*

Proof. Let $D = B \cup \{\omega\}$ be the frame constructed in Proposition 3.30. Then D is compact and KC, but not regular.

By [9, Theorem 9.1.1], every compact strongly Hausdorff frame is regular. Consequently, D is not strongly Hausdorff.

Consider the codiagonal frame morphism

$$\Delta : D \oplus D \longrightarrow D,$$

characterized by

$$\Delta\iota_1 = \mathrm{id}_D = \Delta\iota_2,$$

where ι_1, ι_2 are the canonical coproduct injections.

Since Δ is surjective, the associated nucleus

$$j_\Delta = \Delta_*\Delta$$

determines a quotient of $D \oplus D$ isomorphic to D, and therefore compact.

Suppose that $D \oplus D$ is KC. Then j_Δ must be a closed nucleus. Equivalently, the localic diagonal of D is a closed sublocale, which means that D is strongly Hausdorff. This contradicts Proposition 3.30.

Therefore,

$$D \oplus D \text{ is not KC.}$$

Hence KC is not preserved under finite coproducts. □

We conclude this section by comparing the compact quotients of a frame with those of its quotients. Let $j \in NA$, $F \in A^\wedge$, and $G = j_* F$. Consider the compact quotients

$$A_F = \{x \in A \mid v_F(x) = x\}, \qquad A_{j,G} = \{y \in A_j \mid v_G(y) = y\}.$$

By Lemma 3.24,

$$j \circ v_F \leq v_G \circ j.$$

Since v_G is idempotent, this implies

$$v_G \circ j \circ v_F = v_G \circ j.$$

Thus, the frame morphism $H = v_G \circ j : A \to A_{j,G}$ factors through v_F, inducing

$$h : A_F \longrightarrow A_{j,G}, \qquad h(x) = v_G(j(x)).$$

Proposition 3.32. *The map h is a frame morphism, and the following diagram commutes:*

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A_F \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle h} \\ A_j & \xrightarrow[v_G]{} & A_{j,G}. \end{array} \tag{8}$$

Proof. The map h preserves finite meets and 1, as it is the restriction of $H = v_G \circ j$.

Let $X \subseteq A_F$. Since joins in A_F are computed by applying v_F, we have

$$\begin{aligned} h\left(\bigvee_{A_F} X\right) &= v_G\left(j\left(v_F\left(\bigvee_{A} X\right)\right)\right) \\ &= v_G\left(j\left(\bigvee_{A} X\right)\right) \\ &= \bigvee_{A_{j,G}} h[X]. \end{aligned}$$

Thus h preserves arbitrary joins. Finally,

$$h \circ v_F = v_G \circ j$$

by construction, so the diagram commutes. □

We now specialize the preceding construction to the spatial reflection $\mathcal{U}_A : A \to \mathcal{O}(\operatorname{pt} A)$. Let $F \in A^\wedge$, and let $Q \subseteq \operatorname{pt} A$ be its associated compact saturated subset. By the Hofmann–Mislove correspondence, put $G = \nabla(Q)$.

The point spaces of A_F and $(\mathcal{O}(\operatorname{pt} A))_G$ are both identified with Q. Consequently, the preceding construction yields a commutative diagram relating these compact quotients to $\mathcal{O}Q$.

By [13, Lemma 5.7.1], the point space of the compact quotient A_F is precisely Q. Consequently, the preceding constructions fit into the following commutative diagram:

(9)
$$\begin{array}{ccc} A & \xrightarrow{v_F} & A_F \\ {\scriptstyle \mathcal{U}_A}\downarrow & & \downarrow{\scriptstyle h} \\ \mathcal{O}S & \xrightarrow[v_G]{} & \mathcal{O}S_G \end{array} \qquad \mathcal{U}_{A_F} : A_F \to \mathcal{O}Q, \quad \mathcal{U}_{\mathcal{O}S_G} : \mathcal{O}S_G \to \mathcal{O}Q$$

Since this diagram is determined by $F \in A^\wedge$ and its associated compact saturated subset $Q \in \mathcal{Q}S$, we shall refer to it as the Q-square.

Theorem 3.33. *Under the hypotheses of the Q-square, if A satisfies (H), then*

$$\mathcal{O}Q \cong Q'.$$

Equivalently, the frame of opens of the point space of A_F is isomorphic to a compact closed quotient of a Hausdorff space.

Proof. Suppose that A satisfies (H). Since (H) is conservative, $S = \operatorname{pt} A$ is Hausdorff. Thus Q is closed in S, and hence $Q' \in \mathcal{O}S$.

Since every Hausdorff space is strongly stacked,

$$v_G = [Q'] = u_{Q'}.$$

Therefore,

$$(\mathcal{O}S)_{v_G} = (\mathcal{O}S)_{u_{Q'}} \cong Q'.$$

Moreover, $u_{Q'}$ is a complemented nucleus, so $(\mathcal{O}S)_{v_G}$ is spatial. Since

$$\operatorname{pt}((\mathcal{O}S)_{v_G}) = Q,$$

we conclude that

$$\mathcal{O}Q \cong (\mathcal{O}S)_{v_G} \cong Q'.$$

□

4. Interaction among compact-quotient separation properties

We begin with a result of Sexton [10, Theorem 6.2(a)], which states that, for every topological space S,

$$\mathcal{O}S \text{ neat} \implies S \text{ strongly stacked} \implies S \text{ stacked}.$$

This provides a natural starting point for comparing neatness and strong stackedness in the point-free setting.

Proposition 4.1. *If A is a neat frame, then* $\mathrm{pt}\, A$ *is strongly stacked.*

Proof. Let $S = \mathrm{pt}\, A$. Since neatness is hereditary, $\mathcal{O}S$ is neat. By [10, Theorem 6.2(a)], S is strongly stacked. □

The preceding result concerns the point space of A, rather than A itself. We next identify a condition under which neatness implies strong stackedness of the frame.

Proposition 4.2. *Let A be a neat frame. If*

$$\mathcal{U} \circ v_F = v_G \circ \mathcal{U}$$

for every H–M triple (F, Q, G), then A is strongly stacked.

Proof. Let $S = \mathrm{pt}\, A$ and let (F, Q, G) be an H–M triple. Since A is neat, $\mathcal{O}S$ is neat, and hence S is stacked and packed. Thus Q is closed in S, so $Q' \in \mathcal{O}S$.

By stackedness and Lemma 3.6,

$$v_G(0) = Q'.$$

Since $\mathcal{O}S$ is neat,

$$v_G = u_{Q'} = [Q'].$$

By hypothesis,

$$\mathcal{U} \circ v_F = [Q'] \circ \mathcal{U}.$$

Applying the right adjoint $\mathcal{U}_*$, we obtain

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Therefore A is strongly stacked. □

We next compare KC with the preceding properties.

Proposition 4.3. *Every KC frame is neat.*

Proof. Let $F \in A^\wedge$. Since v_F belongs to the admissibility interval $[v_F, w_F]$ and A is KC, there exists $d \in A$ such that

$$v_F = u_d, \qquad d = v_F(0).$$

Thus A is neat. □

Proposition 4.4. *Let A be a KC frame. Then A is stacked if and only if it is strongly stacked.*

Proof. Since every KC frame is T_1 and neat, for each $F \in A^\wedge$ there exist $d, a \in A$ such that

$$v_F = u_d, \qquad w_F = u_a,$$

where $d = v_F(0)$ and $a = w_F(0)$.

By Proposition 3.11,

$$\begin{aligned} A \text{ stacked} &\iff v_F(0) = w_F(0) \quad \text{for every } F \\ &\iff d = a \quad \text{for every } F \\ &\iff v_F = w_F \quad \text{for every } F \\ &\iff A \text{ strongly stacked.} \end{aligned}$$

□

Proposition 4.5. *Let A be a spatial frame. Then A is KC if and only if it is neat.*

Proof. The forward implication follows from Proposition 4.3. Conversely, suppose that A is spatial and neat. Write $A \cong \mathcal{O}S$ for some topological space S. By the spatial characterization of neatness, S is packed and stacked, so S is strongly stacked. Hence $\mathcal{O}S$ is KC, and therefore so is A. □

Proposition 4.6. *Every neat strongly stacked frame is KC.*

Proof. Let A be neat and strongly stacked. Since neatness implies T_1, by Proposition 3.11,

$$v_F = w_F \qquad (F \in A^\wedge).$$

Moreover, by neatness,

$$v_F = u_{v_F(0)}.$$

Thus every admissibility interval $[v_F, w_F]$ consists of a single closed nucleus. Hence A is KC. □

We have shown that every KC frame is neat and that every neat strongly stacked frame is KC. This naturally raises the question of whether strong stackedness is necessary, or equivalently, whether neatness alone implies the KC property.

An affirmative answer would make neatness and KC equivalent for arbitrary frames, as already happens in the spatial setting. The following example shows that this is not the case.

Theorem 4.7. *There exists a non-spatial neat frame which is neither KC nor strongly stacked. Consequently,*

$$\mathrm{KC} \subsetneq \mathrm{Neat}, \qquad \mathrm{Neat} \not\subseteq \mathrm{S.S.}$$

Proof. Let B be a nontrivial atomless complete Boolean algebra, and consider the frame

$$C = \{(a, b) \in B \times B : a \leq b\},$$

with coordinatewise operations. Let

$$A = C \cup \{\top\},$$

where $\top$ is a new top element, so that $1_C = (1_B, 1_B) < \top$. We shall prove that A is neat but not KC.

We first show that the only Scott-open filter of B is B itself.

Existe alguna propiedad P tal que (H)+P implica staked

T1+S.+P implica (H)?

DONE: Relacionar S. y S.S. con fit y sfit

Relacionar el trabajo con lo presentado por Guram y Sebastian

DONE: Ejemplo de neat no KC

DONE: Comprobar si neat implica stacked

DONE: KC implica S.S.?

Indeed, let F be a Scott-open filter of B. Suppose that $0_B \notin F$. For every $a \in F$, we have $a > 0_B$. Since B is atomless, we can choose b_a such that

$$0_B < b_a < a.$$

In particular, a is not compact. More generally, a complete atomless Boolean algebra has no nonzero compact elements.

Now put

$$d = \bigwedge F.$$

The family

$$D = \{d \vee \neg a : a \in F\}$$

is directed, since F is closed under finite meets. Moreover, by complete distributivity of finite meets over arbitrary joins,

$$\bigvee D = d \vee \bigvee_{a \in F} \neg a = d \vee \neg d = 1_B.$$

Since $1_B \in F$ and F is Scott-open, there exists $a \in F$ such that

$$d \vee \neg a \in F.$$

Consequently,

$$d = a \wedge (d \vee \neg a) \in F.$$

Thus $F = {\uparrow} d$. Scott-openness now implies that d is compact, and hence $d = 0_B$. Therefore $F = B$, proving that

$$B^{\wedge} = \{B\}.$$

We next determine the Scott-open filters of A. Let $G \in A^{\wedge}$.

If $G \cap C \neq \varnothing$, then $1_C \in G$. Consider the frame morphism

$$\delta : B \longrightarrow C, \qquad \delta(b) = (b, b).$$

Since the inclusion $C \hookrightarrow A$ preserves directed joins, the set

$$H = \{b \in B : (b, b) \in G\}$$

is a Scott-open filter of B. Hence $H = B$. In particular, $(0_B, 0_B) \in G$, so $G = A$.

On the other hand, if $G \cap C = \varnothing$, then necessarily $G = \{\top\}$.

Finally, $\{\top\}$ is Scott-open, because $\top$ is compact in A: any family of elements whose join is $\top$ must contain $\top$.

We conclude that

$$A^{\wedge} = \{A, \{\top\}\}.$$

Recall that, for every Scott-open filter F, the associated fitted nucleus is

$$v_F = \left(\bigvee_{a \in F} v_a \right)^{\infty}.$$

Since $0_A \in A$, we have

$$v_A = v_{0_A} = \mathrm{tp} = u_{\top}.$$

Similarly, since $v_\top = \mathrm{id}_A$,

$$v_{\{\top\}} = v_\top = \mathrm{id}_A = u_{0_A}.$$

Thus v_F is a closed nucleus for every $F \in A^\wedge$. Consequently, A is neat.

Let $e = (0_B, 1_B) \in C$. The open nucleus v_e of C is given by

$$v_e(a,b) = e \succ_C (a,b) = (b,b).$$

Indeed, for $(c,d) \in C$,

$$(c,d) \wedge e = (0_B, d),$$

and therefore

$$(c,d) \wedge e \leq (a,b) \quad \iff \quad d \leq b.$$

Define $j : A \to A$ by

$$j(x) = \begin{cases} (b,b), & x = (a,b) \in C, \\ \top, & x = \top. \end{cases}$$

The map j is inflationary and idempotent. Moreover, for $(a,b),(c,d) \in C$,

$$\begin{aligned} j((a,b) \wedge (c,d)) &= j(a \wedge c, b \wedge d) \\ &= (b \wedge d, b \wedge d) \\ &= j(a,b) \wedge j(c,d). \end{aligned}$$

The cases involving $\top$ are immediate. Hence j is a nucleus on A.

Its fixed-point frame is

$$A_j = \{(b,b) : b \in B\} \cup \{\top\}.$$

We claim that A_j is compact. Indeed, suppose that

$$\bigvee_{A_j} x_i = \top$$

for some family $\{x_i\}_{i \in I} \subseteq A_j$. If none of the x_i were equal to $\top$, then $x_i = (b_i, b_i)$ for every i, and hence

$$\begin{aligned} \bigvee_{A_j} x_i &= j\left(\bigvee_A x_i \right) \\ &= j\left(\bigvee_i b_i, \bigvee_i b_i \right) \\ &= \left(\bigvee_i b_i, \bigvee_i b_i \right) \\ &\leq 1_C < \top, \end{aligned}$$

a contradiction. Thus some $x_i = \top$, which proves that A_j is compact.

However,

$$j(0_A) = 0_A,$$

whereas

$$j(0_B, 1_B) = (1_B, 1_B) \neq (0_B, 1_B).$$

Therefore $j \neq \mathrm{id}_A$.

If j were closed, then

$$j = u_{j(0_A)} = u_{0_A} = \mathrm{id}_A,$$

a contradiction. Hence j is not closed.

We have thus constructed a compact quotient A_j of A which is not closed. Therefore A is not KC.

Consequently, there exists a neat frame which is not KC. Since every KC frame is neat, we obtain

$$\mathrm{KC} \subsetneq \mathrm{Neat}.$$

Finally, since every neat strongly stacked frame is KC, the frame A is not strongly stacked either. □

The preceding theorem shows that neatness alone does not imply either KC or strong stackedness. We now examine additional separation conditions under which these implications can be recovered.

Proposition 4.8. *Let A be a KC frame. If A is subfit, then A is strongly stacked.*

Proof. Since A is KC, it is T_1, and both v_F and w_F are closed nuclei for every $F \in A^\wedge$. Thus

$$v_F = u_{v_F(0)} \qquad \text{and} \qquad w_F = u_{w_F(0)}.$$

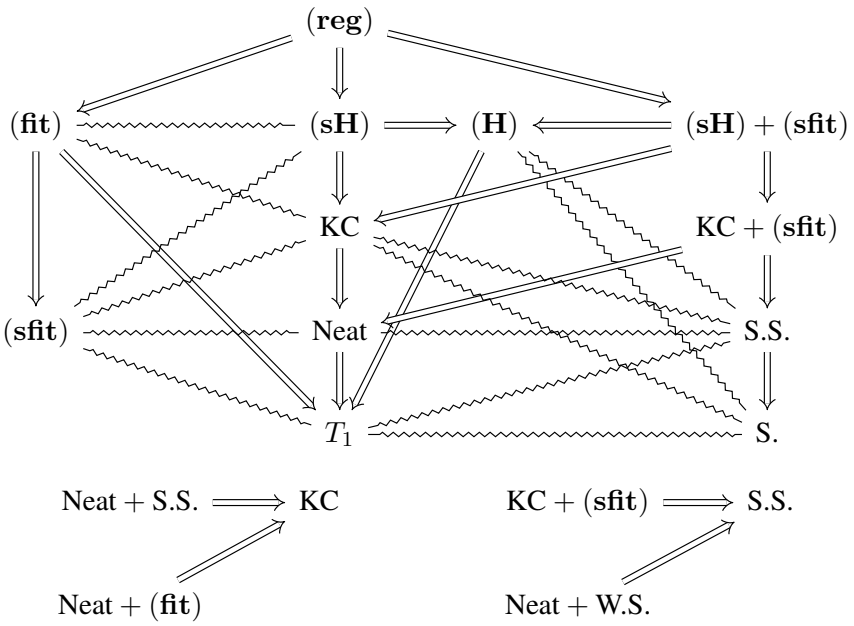
By subfitness and [16, Theorem 3.5], we have $v_F(0) = w_F(0)$. Consequently, $v_F = w_F$ for every $F \in A^\wedge$. By Proposition 3.11, A is strongly stacked. □

Proposition 4.9. *Let A be a frame. If A is neat and fit, then A is KC.*

Proof. Let $j \in NA$ be such that $F = \nabla(j) \in A^\wedge$. Since A is fit, every nucleus is fitted by [16, Theorem 3.4]. Hence $j = v_F$, as v_F is the least nucleus in its admissibility interval.

By neatness, v_F is closed, and therefore so is j. Thus A is KC. □

The following diagram summarizes the relationships among the separation properties considered in this work, including the implications established in [9] and the results obtained here.



5. Relationship with the patch assembly

Example 5.1. Consider the frame $A = [0, 1]$ with its usual order. Its implication is given by

$$a \succ x = \begin{cases} 1, & a \leq x, \\ x, & a > x. \end{cases}$$

Thus the distinguished nuclei satisfy

$$u_a(x) = a \vee x, \qquad v_a(x) = a \succ x.$$

The Scott-open filters of A are

$$A^{\wedge} = \{A\} \cup \{F_m : m \in [0, 1)\}, \qquad F_m = (m, 1].$$

For each $m \in [0, 1)$, the associated fitted nucleus is

$$v_{F_m}(x) = \begin{cases} x, & x \leq m, \\ 1, & x > m, \end{cases}$$

while $v_A = \mathrm{tp}$.

Consequently, the patch assembly PA is generated by the nuclei u_a and $u_a \wedge v_{F_m}$, where

$$(u_a \wedge v_{F_m})(x) = \begin{cases} x, & x \leq m, \\ a \vee x, & x > m. \end{cases}$$

We now identify the spatial counterpart of this construction. The space of points of A is $S = [0, 1)$, with topology

$$\tau = \{\varnothing\} \cup \{[0, a) : 0 < a \leq 1\}.$$

The spatial reflection

$$\mathcal{U}_A : A \longrightarrow \mathcal{O}S, \qquad a \longmapsto [0, a),$$

is an isomorphism.

Since S is totally ordered, [1, Corollary 4.2] implies that $N(\mathcal{O}S)$ is spatial. Moreover, $\mathcal{U}_A$ preserves Scott-open filters. By functoriality of the patch assembly, it induces an isomorphism

$$P(\mathcal{U}_A) : PA \xrightarrow{\cong} P(\mathcal{O}S).$$

The compact saturated subsets of S are precisely the sets

$$Q_m = [0, m], \qquad 0 \leq m < 1,$$

together with $\varnothing$. Their complements are $(m, 1)$. Therefore the patch topology is generated by the sets

$$[0, a) \qquad \text{and} \qquad (m, 1),$$

which generate the usual topology on $[0, 1)$.

Consequently,

$$PA \cong \mathcal{O}([0, 1), \tau_{\text{usual}}) = \mathcal{O}^p S.$$

where ${}^p S$ denotes the patch space of S. In particular, PA is spatial and Hausdorff, although A is not T_1.

Lemma 5.2. *Let A be a frame, if the morphism $P(\mathcal{U})\colon PA \to P\mathcal{O}S$ is an isomorphism, then A is strongly stacked.*

Proof. Consider any open filter F on A and the compact saturated determined by it, say Q, thus we have the nuclei v_F and $\mathcal{U}_*[Q']\mathcal{U} = Q[U]$, we neee to observe the action of $\mathrm{P}(\mathcal{U})$ on these two nuclei, in order to do that remember that we have a conmutative diagram:

$$\begin{array}{ccccc}
A & \longrightarrow & PA & \xhookrightarrow{\iota} & NA \\
{\scriptstyle \mathcal{U}_A}\downarrow & & \downarrow{\scriptstyle P(\mathcal{U}_A)} & & \downarrow{\scriptstyle N(\mathcal{U}_A)} \\
\mathcal{O}S & \longrightarrow & P\mathcal{O}S & \xhookrightarrow{\iota'} & N\mathcal{O}S \\
 & \searrow & \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \sigma} \\
 & & \mathcal{O}^p S & \hookrightarrow & \mathcal{O}^f S
\end{array}$$

since $N(\mathcal{U})\iota = \imath(P\mathcal{U})$, that is, $P(\mathcal{U})$ is the restriction of $N(\mathcal{U})$ to PA, then

$$P(\mathcal{U})(\mathrm{v}_F) = N(\mathcal{U})(\mathrm{v}_F) = \bigvee\{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

since the morphism $\mathcal{U}$ transforms open filters we have

$$\bigvee\{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\} = \bigvee\{u_{v_{\mathcal{U}F}} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

But this representation is the corresponding to the nucleus $\mathrm{v}_{\mathcal{U}F}$. □

Completar esta prueba

Theorem 5.3. *Let A be a T_1 strongly stacked frame. If $S = \mathrm{pt}\, A$ is finitely packed, then A has trivial patch assembly.*

DONE: Considerar incluir información sobre los parches

Proof. Let $F \in A^\wedge$, and let Q be its corresponding compact saturated subset of S. Since S is finitely packed, we may write

$$Q = \{p_1, \dots, p_n\}.$$

As A is T_1 and strongly stacked, we have $v_F = w_F$. Hence, for every $x \in A$,

$$\begin{aligned} v_F(x) &= w_F(x) \\ &= \bigwedge_{i=1}^{n} (p_i \vee x) \\ &= \left(\bigwedge_{i=1}^{n} p_i \right) \vee x. \end{aligned}$$

Thus

$$v_F = u_{\bigwedge_{i=1}^n p_i}.$$

Consequently, every adjusted nucleus v_F is closed. Since PA is generated by nuclei of the form $u_a \wedge v_F$, all its generators are closed.

The closed nuclei form a subframe of NA isomorphic to A. Therefore

$$PA = \{u_a : a \in A\},$$

and hence A has trivial patch assembly. □

Corollary 5.4. *Let A be a T_1 strongly stacked frame. If $S = \mathrm{pt}\, A$ is finitely packed, then A is KC.*

Proof. By Theorem 5.3, every fitted nucleus v_F, with $F \in A^\wedge$, is closed. Moreover, since A is T_1 and strongly stacked, $v_F = w_F$. Thus every admissibility interval $[v_F, w_F]$ consists of a single closed nucleus. Hence A is KC. □

REFERENCES

1. Francisco Ávila, Guram Bezhanishvili, PJ Morandi, and Angel Zaldívar, *The frame of nuclei on an alexandorff space*, Order (2020).
2. Martín H Escardó, *Joins in the frame of nuclei*, Applied Categorical Structures **11** (2003), 117–124.
3. Melvin Hochster, *Prime ideal structure in commutative rings*, Transactions of the American Mathematical Society **142** (1969), 43–60.
4. Karl H Hofmann and Michael W Mislove, *Local compactness and continuous lattices*, Continuous Lattices: Proceedings of the Conference on Topological and Categorical Aspects of Continuous Lattices (Workshop IV) Held at the University of Bremen, Germany, November 9–11, 1979, Springer, 2006, pp. 209–248.
5. P. T. Johnstone, *Stone spaces*, Cambridge Studies in Advanced Mathematics, vol. 3, Cambridge University Press, Cambridge, 1982. MR 698074
6. Peter T Johnstone, *Vietoris locales and localic semilattices*, Continuous lattices and their applications, CRC Press, 2020, pp. 155–180.
7. Jan Paseka and Bohumil Šmarda, *T_2-frames and almost compact frames*, Czechoslovak Mathematical Journal **42** (1992), no. 3, 385–402.
8. J. Picado and A. Pultr, *Frames and locales: Topology without points*, Frontiers in Mathematics, Springer Basel, 2012.
9. Jorge Picado and Aleš Pultr, *Separation in point-free topology*, Springer, 2021.

10. R. A. Sexton and H. Simmons, *An ordinal indexed hierarchy of separation properties*, Topology Proceedings **30** (2006), no. 2, 585–625.
11. RA Sexton and H Simmons, *Point-sensitive and point-free patch constructions*, Journal of Pure and Applied Algebra **207** (2006), no. 2, 433–468.
12. Rosemary A. Sexton, *The use of the frame theoretic assembly as a unifying construct*, Master's thesis, The Victoria University of Manchester, 2000.
13. _____, *A point-free and point-sensitive analysis of the patch assembly*, Ph.D. thesis, The Victoria University of Manchester, 2003.
14. Harold Simmons, *A collection of notes on frames*, Online notes, University of Manchester. Archived author's website, Internet Archive snapshot, July 14, 2019.
15. _____, *The Vietoris modifications of a frame*, Unpublished manuscript, 79 pp. Available via the author's archived website, 2004, Internet Archive snapshot, July 14, 2019.
16. _____, *Regularity, fitness, and the block structure of frames*, Applied Categorical Structures **14** (2006), 1–34.
17. Albert Wilansky, *Between T_1 and T_2*, The American Mathematical Monthly **74** (1967), no. 3, 261–266.